Advanced Microbial Products & Manufacturing Technology

High-CFU Microbial Solutions Engineered for Performance, Stability & Scale

OUR FOCUS

- Converting proven microbial species into stable agricultural products
- Emphasis on Infectivity, pathogenicity, virulence, and shelf-life performance

PARTNERSHIP MODEL

- White-label and private-label manufacturing for agricultural input companies

We are a technology-driven microbial manufacturing partner focused on developing reliable, high-quality microbial products using established microorganisms.

Closing the Biological Performance Gap

WHY MANY MICROBIAL PRODUCTS FAIL

- Rapid CFU decline during storage
- Poor survival under field stress conditions
- Inconsistent performance caused by weak processing and formulation

OUR APPROACH

- Robust manufacturing and stabilization methods that preserve microbial viability
- Viability maintained from production to application
- Consistent field performance across batches

Fungal Biocontrol Portfolio

Fungal Biocontrol Agents with Proven Agricultural Use

CFU Strength: ≥ 1×10^8 – 1×10^9 spores/g (minimum at manufacture)

MICROORGANISMS

- Beauveria bassiana
- Metarhizium anisopliae
- Trichoderma harzianum
- Trichoderma viride
- Lecanicillium lecanii
- Isaria fumosorosea
- Ampelomyces quisqualis
- Paecilomyces lilacinus
- Hirsutella thompsonii

APPLICATIONS

- Insect control
- Disease suppression
- Nematode management
- Rhizosphere health improvement
- Plant growth promotion
- Yield enhancement

Gram-Positive Bacterial Platform

Spore-Forming Bacteria for Stable Agricultural Products

CFU Strength: ≥ 1×10^9 – 1×10^{10} CFU/g (minimum at manufacture)

REPRESENTATIVE SPECIES

- Bacillus subtilis
- Bacillus amyloliquefaciens
- Bacillus velezensis
- Bacillus licheniformis
- Bacillus pumilus
- Bacillus megaterium
- Bacillus firmus
- Bacillus mucilaginosus
- Bacillus mycoides
- Paenibacillus polymyxa

PRIMARY FUNCTIONS

- Disease suppression
- Phosphate and potassium solubilization
- Plant growth promotion
- Induced systemic resistance
- Biocontrol of pathogens
- Soil health improvement

Gram-Negative Bacterial Platform

High-Activity Rhizosphere Bacteria

CFU Strength: ≥ 1×10^8 – 1×10^9 CFU/g (minimum at manufacture)

REPRESENTATIVE SPECIES

- Pseudomonas spp.
- Pseudomonas fluorescens
- Pseudomonas striata
- Azotobacter chroococcum
- Azospirillum brasilense
- Azospirillum lipoferum
- Rhizobium spp.
- Enterobacter cloacae
- Frateuria aurantia
- Acidithiobacillus spp.

PRIMARY FUNCTIONS

- Nitrogen fixation
- Nutrient solubilization
- Root-zone colonization
- Siderophore production
- Phytohormone synthesis
- Biocontrol activity

Solid State Fermentation (SSF)

Manufacturing Platform for Fungal Products

Biological Alignment

- Growth conditions aligned with natural fungal biology
- Mimics natural substrate colonization
- Optimal for spore development

Product Quality

- Production of robust aerial conidia
- High spore concentration per gram
- Superior spore viability

Field Performance

- Improved tolerance to UV exposure
- Enhanced desiccation resistance
- Better survival in field conditions

Liquid Fermentation & Spray Drying

Manufacturing Platform for Gram-Positive Bacteria

PROCESS

- Suitable for spore-forming bacterial species
- High-density biomass production
- Precise control of fermentation parameters

DRYING TECHNOLOGY

- Controlled drying for moisture uniformity
- Consistent particle size distribution
- Optimized for stability

QUALITY OUTPUT

- Consistent CFU delivery
- Excellent shelf stability
- Batch-to-batch consistency

Lyophilization (Freeze Drying)

Manufacturing Platform for Gram-Negative Bacteria

GENTLE PROCESSING

- Gentle dehydration without thermal stress
- Preserves cellular integrity
- Minimal damage to sensitive cells

VIABILITY MAINTENANCE

- Maintains viability of non-spore-forming bacteria
- High survival rates post-processing
- Long-term stability

FIELD APPLICATION

- Rapid solubility and activation
- Easy to use in field conditions
- Improved stability vs liquid products

Manufacturing Platforms Aligned with Microbial Biology

FUNGI

Solid State Fermentation

- Substrate-based growth systems
- Production of aerial conidia
- High spore density & environmental resilience
- Optimized for field survival

GRAM-POSITIVE

Liquid Fermentation + Spray Drying

- High-density cultivation
- Controlled drying for stability
- CFU consistency across batches
- Excellent shelf-life properties

GRAM-NEGATIVE

Lyophilization

- Heat-free dehydration process
- Preserves sensitive cells
- Fast reactivation and ease of use
- Superior stability for non-spore formers

Key Takeaway: Manufacturing technology is selected based on microbial biology to ensure stability and consistent product quality.

Formulation & Delivery Formats

Designed for Ease of Use and Compatibility

Water Soluble Powder (WSP)

- Uniform dispersion in tanks
- Low dust & safe handling
- Easy to measure and apply

Tablet Formulations

- Pre-measured accurate dosing
- Suitable for precision agriculture
- Extended shelf life

Oil Dispersion (OD)

- Enhanced leaf adhesion
- Reduced evaporation losses
- Better UV protection

APPLICATION COMPATIBILITY

- Seed treatment applications
- Foliar spray applications
- Soil drench and incorporation

Your Brand, Supported by Reliable Manufacturing

WHAT WE DELIVER

- High and consistent CFU levels at the time of supply
- Products with improved shelf-life under ambient storage
- Uniform product quality across production batches
- Technical documentation and support
- Custom formulation development
- Scale-up and production planning

COMMERCIAL IMPACT

- Greater confidence in product quality
- Improved consistency in field results
- Stronger credibility with distributors and growers
- Reduced product returns and complaints
- Enhanced brand reputation in market
- Competitive advantage through quality